Structural Break Detection in the Aerospace Sector

Modélisation Statistique — Séries Temporelles ENSIIE Paris-Évry · Master 1 · 2026

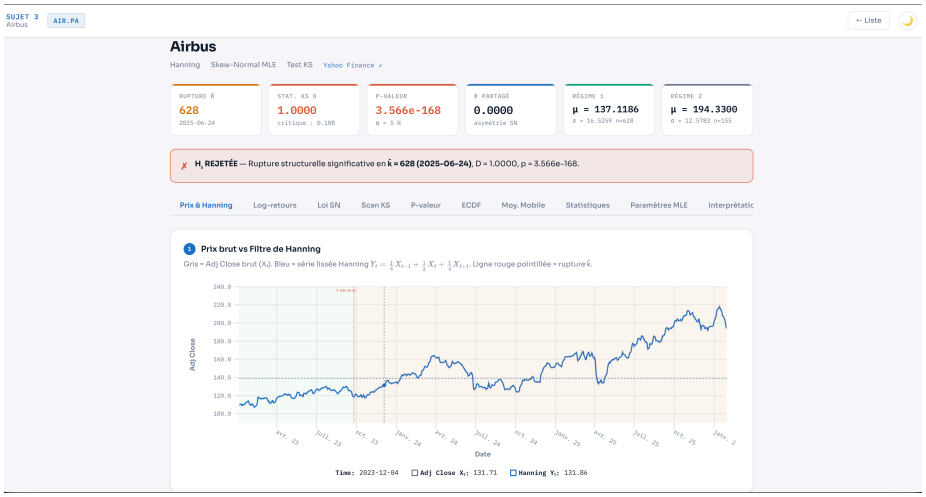

Figure 1: Web App Preview

Group 2 — Members

Role	Name
Coordinator	SEK Sopheak Voatei
Member	THONG Ousaphea
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Project Overview

This project detects **structural breaks** (regime changes) in the daily adjusted closing prices of **10 major aerospace & defence companies** using a combination of:

1. **Hanning filter** — smooths the raw price series

2. **KS scan** — finds the most likely breakpoint nonparametrically
3. **Skew-Normal MLE** — fits a distribution to each regime
4. **Flask web app** — interactive dashboard to explore results

Companies Analysed

Ticker	Company
AIR.PA	Airbus SE
AM.PA	Dassault Aviation
BA	The Boeing Company
BAESY	BAE Systems plc
BDRAF	Bombardier Inc.
GD	General Dynamics Corporation
HON	Honeywell International Inc.
LMT	Lockheed Martin Corporation
NOC	Northrop Grumman Corporation
RTX	RTX Corporation

Methodology

1. Hanning Filter

Raw prices X_t are smoothed to reduce micro-structure noise:

$$Y_t = \frac{1}{4}X_{t-1} + \frac{1}{2}X_t + \frac{1}{4}X_{t+1}$$

2. Breakpoint Model

The filtered series follows a piecewise Skew-Normal model with unknown breakpoint k:

$$Y_t \sim \begin{cases} \mathrm{SN}(\mu_1, \sigma_1, \theta) & \text{if } t \leq k \\ \mathrm{SN}(\mu_2, \sigma_2, \theta) & \text{if } t > k \end{cases}$$

where the Skew-Normal density is:

$$f(y;\, \mu, \sigma, \theta) = \frac{2}{\sigma}\, \varphi\Big(\frac{y-\mu}{\sigma}\Big)\, \Phi\Big(\theta \cdot \frac{y-\mu}{\sigma}\Big)$$

with φ the standard Normal pdf and Φ the standard Normal cdf.

3. Breakpoint Estimation

The breakpoint is estimated by maximising the two-sample Kolmogorov–Smirnov statistic over all candidate positions:

$$\hat{k} = \arg \max_{k \in \{30, \ldots, n-30\}} D_{k,\, n-k}$$

where $D_{k,\, n-k} = \sup_x |\hat{F}_1(x) - \hat{F}_2(x)|$.

4. Hypothesis Test (KS homogeneity)

- **H** : no break — both segments follow the same distribution
- **H** : break exists — segments follow different distributions
- Significance level: $\alpha = 0.05$

5. MLE Estimation

Parameters $(\mu_1, \sigma_1, \mu_2, \sigma_2, \theta)$ are estimated jointly by maximising the log-likelihood:

$$\ell = \sum_{t=1}^{\hat{k}} \log f(Y_t;\, \mu_1, \sigma_1, \theta) + \sum_{t=\hat{k}+1}^{n} \log f(Y_t;\, \mu_2, \sigma_2, \theta)$$

via the **L-BFGS-B** numerical optimiser.

Project Structure

```
.
  app.py                  # Flask web application
  Lib_Aerospace.py        # Core library (filter, KS scan, MLE, plots)
  Dic_Aerospace.py        # Ticker → company name dictionary
  data/
     Secteur_Aerospace/  # CSV price data (one file per ticker)
  results/
     json/               # Cached analysis results
     notebook/           # Member Jupyter notebooks
  report/
     report.tex          # LaTeX report
     generate_figures.py # Script to generate all report figures
     figures/            # Output PNGs (generated)
  static/                 # CSS / JS for the web app
  templates/              # HTML templates
  pyproject.toml
```

Installation

Requirements

- Python 3.12
- uv (recommended) or pip

With uv (recommended)

```
# Clone the repo
git clone https://github.com/SokhengDin/Projet-Series-Temporelle-S4.git
cd Projet-Series-Temporelle-S4

# Create environment and install dependencies
uv sync
```

With pip

```
pip install flask matplotlib numpy pandas scipy statsmodels
```

Running the Web App

```
# With uv
uv run python app.py

# Or with plain Python
python app.py
```

The app opens automatically at http://127.0.0.1:8000.

Select a company from the dropdown to see: - Raw and Hanning-filtered price series - KS scan with estimated breakpoint - Regime histograms with fitted Skew-Normal densities - MLE parameter estimates and test verdict

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Results Summary

The KS test **rejects H for all 10 companies** at the 5% level. Every stock underwent at least one distributional regime change since January 2023, with higher mean prices and increased dispersion in the post-break regime.

Ticker	Break date	KS stat	Decision
AIR.PA	2025-06-24	1.0000	Reject H

Ticker	Break date	KS stat	Decision
AM.PA	2025-01-21	1.0000	Reject H
BA	2025-12-15	0.6436	Reject H
BAESY	2023-02-17	1.0000	Reject H
BDRAF	2024-05-15	1.0000	Reject H
GD	2023-11-13	1.0000	Reject H
HON	2024-06-04	0.8634	Reject H
LMT	2025-12-11	0.7935	Reject H
NOC	2025-07-22	1.0000	Reject H
RTX	2024-07-25	1.0000	Reject H

If you encounter any issues running the application, feel free to contact soknan.koum@ensiie.eu, veasna.ra@ensiie.eu and sokheng.din@ensiie.eu